

CONTACT INFORMATION	Department of Industrial & Systems Engineering, Viterbi School of Engineering 3715 McClintock Avenue Los Angeles, CA, 90089-0193	Website febattista.github.io Email fb_237@usc.edu
RESEARCH INTERESTS	Discrete optimization with contributions spanning theory, methodology, and computation toward more effective solution approaches for complex mixed-integer linear problems and the development of open-source tools for the community. Combinatorial optimization, particularly the design of strong relaxations for graph problems, such as, the maximum clique/independent set and graph coloring.	
ACADEMIC EXPERIENCE	University of Southern California, Dept. of Industrial and Systems Engineering, Los Angeles, CA, USA Postdoctoral Researcher under Karmel S. Shehadeh <i>February 2026 – Present</i> Lehigh University, Dept. of Industrial and Systems Engineering, Bethlehem, PA, USA Postdoctoral Researcher under Ted Ralphs <i>February 2023 – January 2026</i>	
EDUCATION	Università di Roma Sapienza Dept. of Computer, Control, and Management Engineering, Rome, Italy Ph.D in Operations Research <i>November 2019 – January 2023</i> Thesis: On Semidefinite Lift-and-Project of Combinatorial Optimization Problems Advisors: Marianna De Santis, Fabrizio Rossi, Stefano Smriglio Alpen-Adria-Universität Institut für Mathematik, Klagenfurt, Austria Visiting Ph.D student under Angelika Wiegele <i>February 2022 – July 2022</i> Università degli Studi dell'Aquila Dept. of Inf. Eng., Computer Science and Mathematics, L'Aquila, Italy Master's degree in Computer Science, <i>Magna Cum Laude</i> <i>July 2019</i> Bachelor's degree in Computer Science, <i>Magna Cum Laude</i> <i>October 2017</i>	
REFEREED PUBLICATIONS	[1] F. Battista, F. Rossi, and S. Smriglio. <i>Application of the Lovász-Schrijver Operator to Compact Stable Set Integer Programs</i>. In: <i>Mathematical Programming Computation</i> (2025). DOI: 10.1007/s12532-025-00298-8. [2] F. Battista and M. De Santis. <i>Dealing with Inequality Constraints in Large-Scale Semidefinite Relaxations for Graph Coloring and Maximum Clique Problems</i>. In: <i>4OR. A Quarterly Journal of Operations Research</i> (2024), pp. 1–31. DOI: 10.1007/s10288-024-00569-5. [3] F. Battista. <i>On Semidefinite Lift-and-Project of Combinatorial Optimization Problems</i>. PhD thesis. Università di Roma Sapienza, 2023. HDL: 11573/1668673. URL: https://hdl.handle.net/11573/1668673.	
WORKING PAPERS	[4] F. Battista and T. K. Ralphs. <i>Improving Directions in Mixed Integer Bilevel Linear Optimization</i>. arXiv preprint. COR@L Laboratory, Lehigh University, 2025. URL: https://doi.org/10.48550/arXiv.2511.03566. <i>Submitted</i>. [5] S. Fallah, F. Battista, and T. K. Ralphs. <i>A Branch-and-Bound Algorithm for Multiobjective Mixed Integer Linear Optimization</i>. Working paper. COR@L Laboratory, Lehigh University, 2025.	

SOFTWARE

- [6] F. **Battista**, F. Rossi, and S. Smriglio. *SDP_lift_and_project* (2025). URL: https://github.com/febattista/SDP_lift_and_project. *Maintainer*.
- [7] F. **Battista** and M. De Santis. *ADAL-ineq* (2024). URL: <https://github.com/febattista/ADAL-ineq>. *Maintainer*.
- [8] S. DeNegre, T. Ralphs, and S. Tahernejad. *MibS Version 1.2* (2024). URL: <https://github.com/coin-or/MibS>. *Contributor*.
- [9] T. Ralphs, M. Güzelsoy, and A. Mahajan. *SYMPHONY Version 5.7* (2023). URL: <https://github.com/coin-or/SYMPHONY/>. *Contributor*.

STUDENTS

Doctoral Students Mentorship

- Shannon Kelley, Lehigh University, ISE, Advisor: Ted Ralphs 2023 – 2026
- Samira Fallah, Lehigh University, ISE, Advisor: Ted Ralphs 2023 – 2026
- Hesam Shaelaie, Lehigh University, ISE, Advisor: Ted Ralphs 2023 – 2026
- Suresh Bolusani, Lehigh University, ISE, Advisor: Ted Ralphs 2023 – 2026

TEACHING EXPERIENCE

Teaching Assistant

- *ISE 418* Integer Programming, Ph.D. course, Lehigh University 2024
- Operations Research Laboratory, B.Sc. course, Università Roma Sapienza 2020
- Advanced Calculus, B.Sc. course, Università Roma Sapienza 2020

TALKS

- [1] F. **Battista**, T. K. Ralphs, and S. Fallah. *The Restricted Value Function of MILPs and Its Construction with SYMPHONY*. In: *18th INFORMS Computing Society (ICS) Conference* (2025).
- [2] F. **Battista** and T. K. Ralphs. *A Branch-and-Cut Algorithm for Mixed-Integer Bilevel Linear Optimization Based on Improving Directions*. In: *International Symposium on Mathematical Programming* (2024).
- [3] F. **Battista** and T. K. Ralphs. *Exploiting Dual Functions in Mixed Integer Bilevel Linear Programs*. In: *INFORMS Annual Meeting* (2023).
- [4] F. **Battista**, F. Rossi, and S. Smriglio. *Application of the Lovász-Schrijver Operator to Representative Formulation for Graph Coloring Problem*. In: *International Symposium on Combinatorial Optimization* (2022).
- [5] F. **Battista** et al. *On Semidefinite Lift-and-Project Relaxations for Combinatorial Optimization Problems*. In: *University of Pavia* (2022).
- [6] F. **Battista** and M. De Santis. *Dealing with Inequalities in Large Scale Semidefinite Programs*. In: *International Conference on Optimization and Decision Science* (2021).

SERVICE

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